Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview across primary energy, electricity generation, and technology pathways.

~14%

Primary energy from renewables

2024, substitution method

~30%

Electricity from renewables

2025 global average

Source: Our World in Data – Renewable Energy

75% of Global GHG Emissions Still Come from Fossil Fuels

The renewable transition is both a climate and public-health imperative.

75%

of global greenhouse gas emissions

come from burning fossil fuels for energy.

5M+

premature deaths annually

linked to local air pollution from fossil fuel combustion.

2

low-carbon options

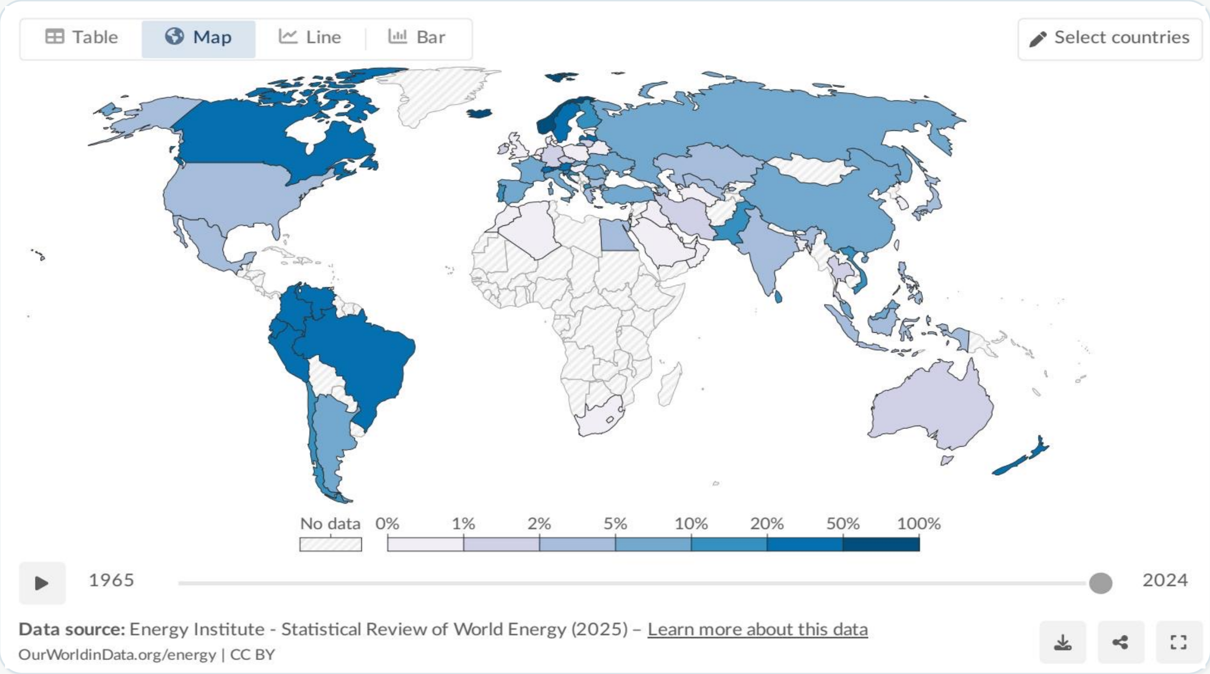
renewable technologies and nuclear energy can reduce CO_2 and air pollution.

! Core problem: energy systems remain fossil-fuel dominated, while electricity is the sector where renewables have gained the most ground.

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

Renewables Now Supply ~14% of Global Primary Energy

Primary energy includes electricity, transport, and heating combined.



Share of primary energy from renewables by country, 2024

~one-seventh

of global primary energy comes from renewable technologies.

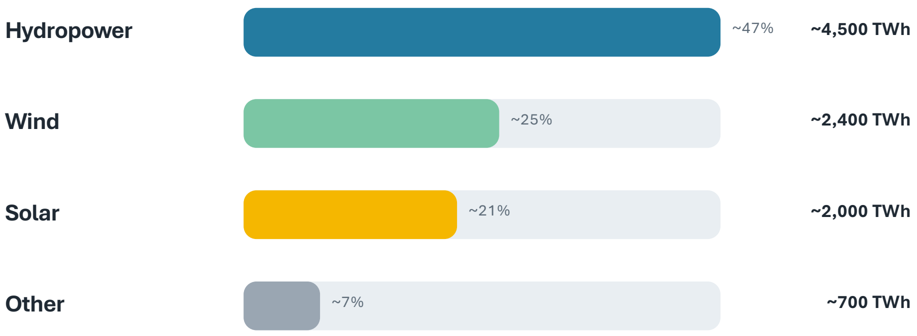
2024 • substitution method

- Transport and heating remain harder to decarbonize than power.
- Norway, Brazil, and New Zealand show large renewable shares.
- Middle Eastern fossil-fuel producers and parts of Asia remain below 5%.

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Hydropower still leads, but wind and solar combined now nearly match its output.

Approx. global renewable generation by technology, 2024



TWh and share of renewable electricity

What is changing fastest?

- Solar has increased dramatically since 2010.
- Falling costs and policy support are key drivers.
- Wind and solar are the fastest-growing energy technologies in history.

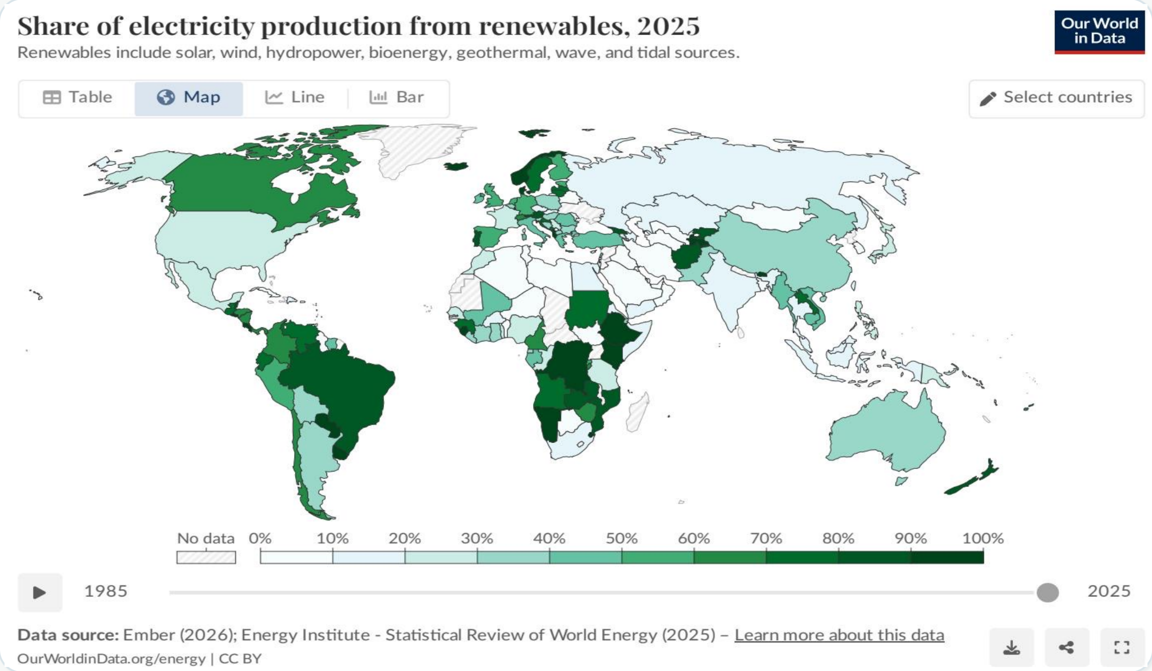
Wind + solar: ~4,400 TWh

nearly matching hydropower's ~4,500 TWh

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

Almost One-Third of Global Electricity Now Comes from Renewables

Electricity is where the renewable transition has advanced furthest.



Share of electricity from renewables by country, 2025

Selected renewable electricity shares

Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

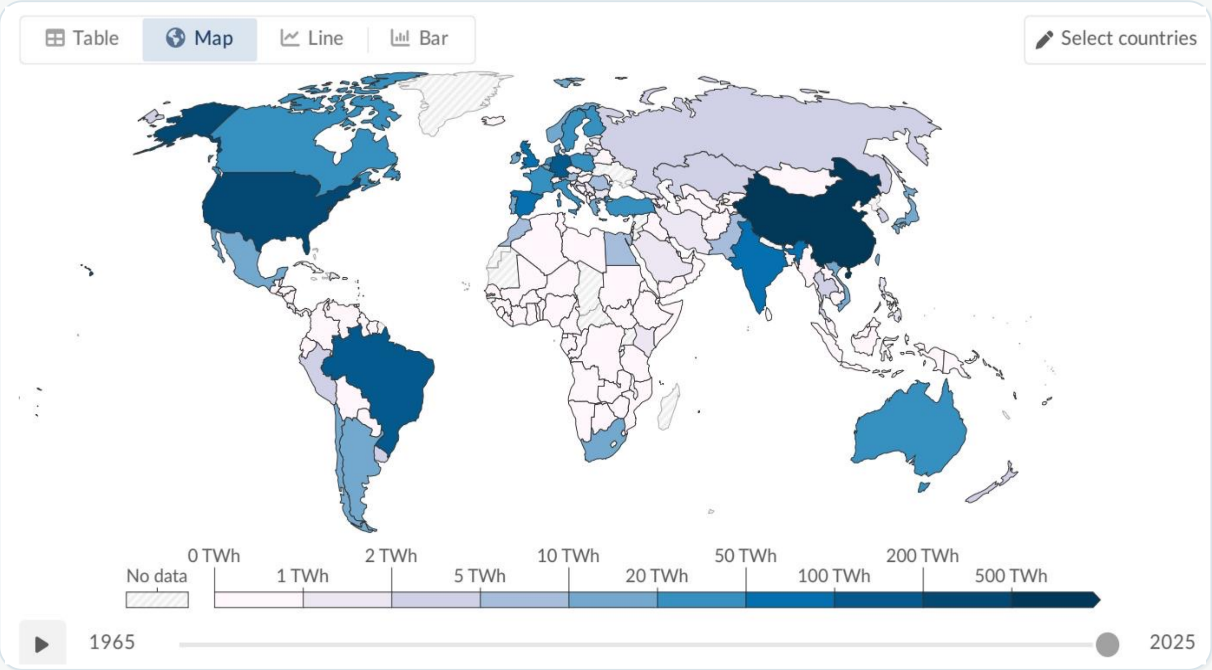
World average ~30%

Leaders reach 60–90%+ renewable electricity, while several African and Middle Eastern countries remain below 10%.

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

Hydropower Remains the Largest Renewable Source at ~50% of Generation

A century-old technology still anchors renewable electricity in many regions.



Hydropower generation by country, 2025

~4,500 TWh

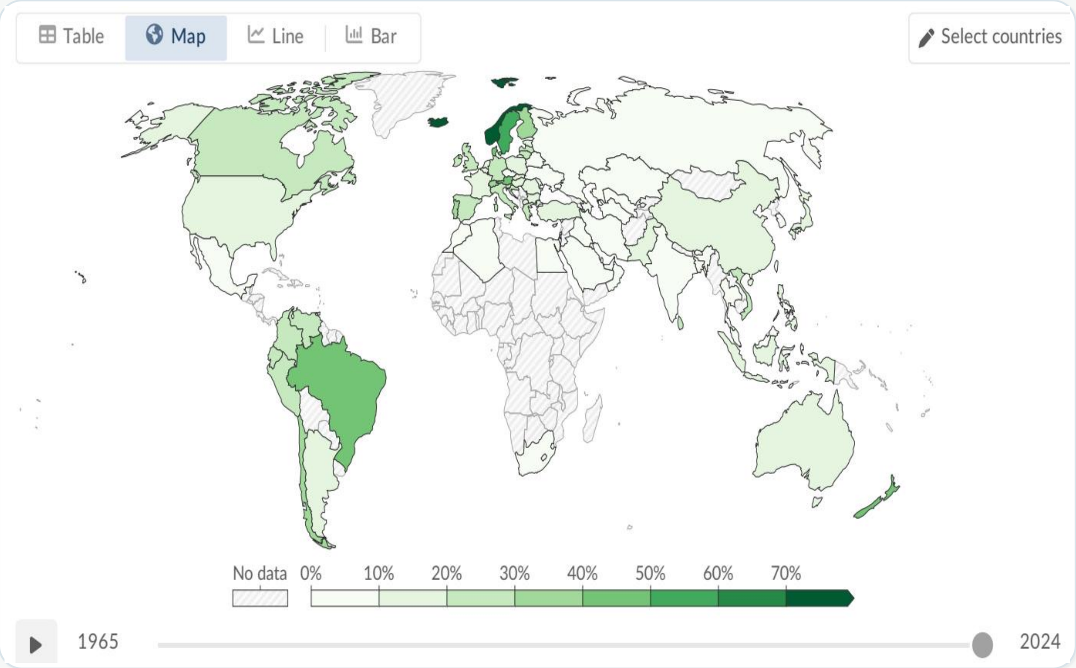
global hydropower generation

- Roughly half of all renewable generation.
- Top producers include China, Brazil, and Canada.
- Growth potential is more limited than wind and solar.

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

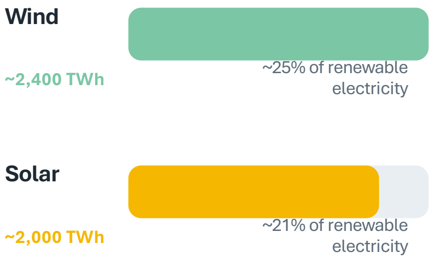
Wind and Solar Are the Fastest-Growing Energy Technologies in History

Generation—not installed capacity—shows both technologies are now major contributors.



Wind power generation by country, 2025

Wind vs. solar generation

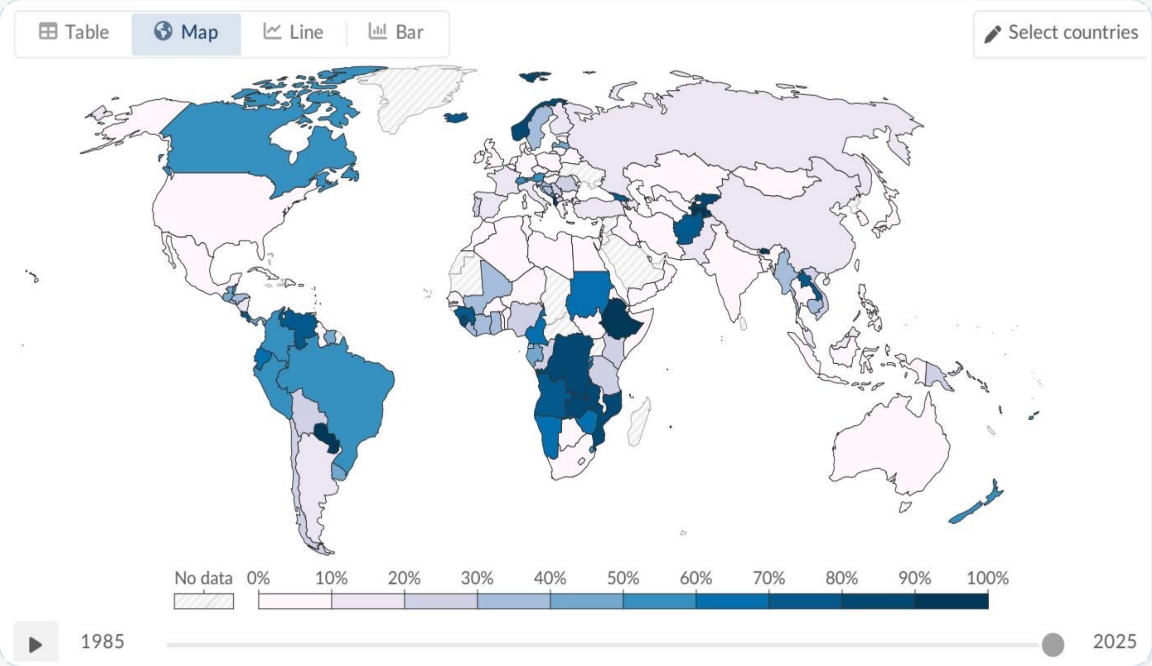


Solar is the fastest-growing source, driven by falling costs and policy support.

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

Resource endowments, infrastructure, and investment shape uneven transition pathways.



Share of electricity from hydropower by country, 2025

Transition geography

Leaders

Scandinavia, Brazil, and Canada benefit from abundant hydropower.

Wind + solar investors

Denmark and Germany have invested heavily in wind and solar.

Lagging regions

Sub-Saharan Africa has minimal infrastructure; many Middle Eastern nations remain fossil-fuel dependent.

Below 10% in several African & Middle Eastern countries

Source: Our World in Data – Renewable Energy; Ember (2026); Energy Institute (2025)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

The world has crossed important electricity milestones, but economy-wide energy remains far from decarbonized.

~14% / ~30%	Renewables supply ~14% of global primary energy and ~30% of electricity.
Hydro anchors	Hydropower still generates roughly half of renewable electricity.
Wind + solar catch up	Together they now produce nearly as much electricity as hydropower.
Regions diverge	Scandinavia and Latin America lead; much of the Middle East and Sub-Saharan Africa lag.
Beyond power	Transport and heating lag behind electricity, holding down the total energy share.

Policy implication: accelerate renewable deployment while extending decarbonization beyond electricity into transport and heating.

Source: Our World in Data – Renewable Energy